Mahesh Gupta Kesarla

Principal Architect —Platform Architecture and AI Driven Modernization

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Executive Summary

Principal Architect with 22 years of deep experience in **building enterprise platforms and modernizing legacy systems** using cloud-native and AI-driven architectures. Active member of Enterprise Architecture and AI Solution Architecture Boards, shaping reference architectures, modernization standards, and AI adoption patterns.

Hands-on in systems design, multi-agent LLM solutions, and platform modernization, while steering governance, technical strategy, and cross-team execution across global teams.

Core Skills

• Solution Architecture • Microservices • Cloud-Native • Legacy Refactoring • LLMs (GPT, Claude, Gemini) • Multi-Agent Systems • RAG • LangChain/LangGraph • AI Governance • AWS • Serverless • Datadog • API Design • Messaging • DevSecOps • Java • Python • Technical Strategy • Global Delivery

Strategic Programs

eServices - Global Service Shop Platform

RMD - Remote Monitoring & Diagnostics

• Platform supports 8,000 global users across 180+ service shops enabling $700M annual material transactions with 24×7 availability.
• Enhanced SmartShop lean work orders to improve high value parts life by 20–40%
• LHFM (Locos Held for Material): reduced material wait time by 15% for CSX.
• Release window reduced from 6 hours → 10 minutes.
• Architecture: ~50 microservices (Java/Spring Boot/Python), Angular/React UIs, AWS (EC2, RDS/Postgres, Lambda, S3, SQS, Redis, Transcribe, Translate), IBM MQ, Datadog, Grafana.
• Real-time health, diagnostics, and reliability metrics for ~30,000 locomotives.

TechPorter - AI Multi-Agent Modernization Platform

• Designed and built an agentic AI solution to accelerate modernization, using multi-model orchestration, hierarchical prompting and context-aware memory.
• Developed a multi-agent system (Documentation Agent, Migration Agent, Security Agent, Judge Agent) that interprets legacy code, transforms and validates through shared context.
• Modernized 8 apps with 80% automation, reducing effort from 6 months to 1 week
• Used Hybrid-LLM routing across Gemini, Claude Sonnet, GPT models and spec-driven frameworks.
• Upgrades: Java 8 → Java 17, Spring → Spring Boot, cloud-ready refactoring.

Genie - AI Assistant for Service Shop Technicians

• Agentic RAG system powering guidance from 3,000 PDFs (multi-version, multi-language).
• Improved compliance adherence by 20% during pilot.
• Stack: LangChain, LangGraph, Python, vector search, Claude Opus, Sonnet, auth & audit.

Architecture & AI Leadership

• Defined and published reference architectures, modernization patterns, and enterprise standards for core platforms and AI-driven solutions
• Guide teams toward target-state architectures and ensure alignment with enterprise strategy, standards using decision trees and maturity models.
• Mentor 4–5 people annually (intern to architect), developing soft-skills, critical thinking, and next-role readiness—while cultivating a no-hierarchy, ownership-driven culture.
• Evaluate AI/GenAI solutions and ensure they meet enterprise standards for security, scalability, and responsible use.
• Provide architectural oversight for AI initiatives across global business units.

Technical Leadership

• Defined secure SDLC aligned with ISO 27001 including environment isolation, secure testing, secure coding, vulnerability management.
• Led 4 (2 external + 2 internal) audits across service shops in Paris and the US.
• Dotted-line leadership for 5 Architects and 4 Engineering Managers; indirect responsibility for 50 FTE and 30 contractors across US/Mexico/India.
• Ownership from requirements → production deployment; preventive maintenance strategy, error-pattern extraction, modernization and security standards.
• Influence $3M annual IT budget; define technology roadmaps, innovation themes, and tech debt elimination plans.

Certifications

• TOGAF 9 Certification (The Open Group)
• AWS AI Practitioner (Early Adopter)
• AWS Solutions Architect - Associate (AWS Certified)
• Oracle Certified Professional - Java SE Programmer
• Oracle Certified Expert – Java EE Web Component Developer
• IBM Cloud Application Developer

Professional Experience

Principal Architect — GE Transportation / Wabtec *(Aug 2017 – Present)*
Own tech strategy and innovation, make architecture decisions, and ensure smooth platform operations
Technical Architect — IBM India Pvt Ltd *(Sep 2004 – Aug 2017)*
Led application architecture and design of large-scale applications (IBM MobileFirst for iOS, TD Bank MBNA MDM, Bank of America Online Banking, IBM MSS) across regulated environments.
Software Programmer — HP India Pvt Ltd *(Sep 2003 – Sep 2004)*
Developed features and resolved defects for content delivery system used by Starbucks and Walmart.

Education

Master of Computer Applications (MCA)